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(1) Stationary ARMA processes

A time series Y_t is governed by the following AR(2) process.

$$(1 - 1.1B + 0.18B^2)Y_t = \varepsilon_t \quad \text{for} \quad E(\varepsilon_t \varepsilon_\tau) = \begin{cases} 1 & (t = \tau) \\ 0 & (t \neq \tau) \end{cases}$$

Here B is the backward shift operator and ε_t is noise.

- Is it covariance-stationary? Why? (*Hint: refer to p.28 of the lecture on Time Series in Complex Systems.*)
- If so, calculate its autocovariance. (*Hint: refer to p.38 of the lecture on Time Series in Complex Systems.*)

(2) Linear regression models

You have a time series X_t which has a large number of data points. Let $\Delta X_t \equiv X_t - X_{t-1}$ be its backward difference.

- You have first fitted the following model

$$\Delta X_t = +0.51(\pm 0.29) - 0.11(\pm 0.04)X_{t-1} - 0.39(\pm 0.09)\Delta X_{t-1} - 0.32(\pm 0.09)\Delta X_{t-2} - 0.22(\pm 0.10)\Delta X_{t-3}$$

where the number after $\pm$ is the standard error of its preceding coefficient. Now you want to perform a unit root test.

- What is the t -statistic of the lagged term X_{t-1} ? At what significance level can you reject the null hypothesis? (*Hint: use the ADF table provided at the end of this problem set.*)
 - What can you conclude if the null hypothesis is rejected?
- After that, you have fitted three other models

$$\Delta X_t = +0.002(\pm 0.014) - 0.31(\pm 0.10)\Delta X_{t-1} \quad (1)$$

$$\Delta X_t = +0.021(\pm 0.158) - 0.46(\pm 0.10)\Delta X_{t-1} - 0.39(\pm 0.11)\Delta X_{t-2} - 0.25(\pm 0.08)\Delta X_{t-3} \quad (2)$$

$$\Delta X_t = +1.279(\pm 0.57) - 0.51(\pm 0.10)\Delta X_{t-1} - 0.44(\pm 0.11)\Delta X_{t-2} - 0.30(\pm 0.09)\Delta X_{t-3} - 0.16(\pm 0.07)Y_{t-1} \quad (3)$$

where Y is an independent variable.

- (i) What is the t -statistic of Y_{t-1} ? At what significance level can you reject the null hypothesis for Y_{t-1} ? (Hint: use a t -distribution table.)
- (ii) You want to use models (1) to (3) and the following information to forecast X_5 . Calculate their respective predicted values.

t	X_t	Y_t
1	0.8	7.7
2	4.3	7.9
3	2.9	7.7
4	1.4	7.0

(3) Fourier Transform of Impulses

In this question, you will learn about the Fourier Transform of impulses. Use the notations of Fourier Transform on p.58 of the lecture on Time Series in Complex Systems.

- (a) Let us begin by considering a Gaussian impulse

$$f(t) = ce^{-b(t-t_0)^2}$$

where t is the time variable and (b, c, t_0) are constants. What is its Fourier transform?

- (b) A Dirac delta impulse $\delta(t - t_0)$ can be approximated by using the Gaussian impulse above and setting $b = 1/(2\sigma^2)$ and $c = 1/\sqrt{2\pi\sigma^2}$.

- (i) A Dirac delta function $\delta(x)$ has the properties that

$$\delta(x) = \begin{cases} 0 & (x \neq 0) \\ \infty & (x = 0) \end{cases} \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(x) dx = 1$$

Describe how the Gaussian impulse satisfies them as $\sigma \rightarrow 0$.

- (ii) Using this approximation and the result in (a), what is the Fourier transform of a Dirac delta impulse?
- (c) Next, we are going to consider some impulses that have temporal power law decay. Let us begin by considering a step function

$$f(t) = \begin{cases} 1 & (t > 0) \\ 0 & (t < 0) \end{cases}$$

If you naively perform Fourier transform on it, you will get something that is undefined. Instead, you can first perform Fourier transform on

$$f(t) = \begin{cases} e^{-\alpha t} & (t > 0) \\ 0 & (t < 0) \end{cases}$$

and then let $\alpha \rightarrow 0$ to retrieve the Fourier transform of the step function.

- (i) Hence, what is the Fourier Transform of the step function? (*Hint: its discontinuity at $t = 0$ will give a delta function in your answer.*)
- (ii) What is its power spectral distribution function? Comment on your result.
- (d) Using the method shown in (c), what is the Fourier transform of the following sign function?

$$f(t) = \begin{cases} 1 & (t > 0) \\ -1 & (t < 0) \end{cases}$$

- (e) Using the result in (d), what are the Fourier transforms of (i) $f(t) = 1/t$, (ii) $f(t) = |t|$, and (iii) $1/t^2$ for $-\infty \leq t \leq \infty$? [*Hint: for $f(t) = 1/t$, start from its inverse Fourier transform and redefine the parameters in the integral, i.e. $t \rightarrow -\omega$ and $\omega \rightarrow t'$.*]

Augmented Fuller-Dickey Table

Significance level	0.01	0.025	0.05	0.10
Sample Size T	The τ statistic: No Constant or Time Trend ($a_0 = a_2 = 0$)			
25	-2.65	-2.26	-1.95	-1.60
50	-2.62	-2.25	-1.95	-1.61
100	-2.60	-2.24	-1.95	-1.61
250	-2.58	-2.24	-1.95	-1.62
300	-2.58	-2.23	-1.95	-1.62
∞	-2.58	-2.23	-1.95	-1.62
	The τ_μ statistic: Constant but No Time Trend ($a_2 = 0$)			
25	-3.75	-3.33	-2.99	-2.62
50	-3.59	-3.22	-2.93	-2.60
100	-3.50	-3.17	-2.89	-2.59
250	-3.45	-3.14	-2.88	-2.58
500	-3.44	-3.13	-2.87	-2.57
∞	-3.42	-3.12	-2.86	-2.57
	The τ_τ statistic: Constant + Time Trend			
25	-4.38	-3.95	-3.60	-3.24
50	-4.15	-3.80	-3.50	-3.18
100	-4.05	-3.73	-3.45	-3.15
250	-3.99	-3.69	-3.43	-3.13
500	-3.97	-3.67	-3.42	-3.13
∞	-3.96	-3.67	-3.41	-3.12

Source: The table is reproduced from Fuller (1996).